

by mathematical and scientific foundations; rather it's a practice with a set of guidelines to craft precise, concise, creative wording of text to instruct an LLM to carry out a task. Another way to put it is that "prompt engineering is the art of communicating with a generative large language model." [2]

To communicate with LLM with precise and task-specific instructions prompts, Bsharat et.al [3] present a comprehensive principled instructions and guidelines to improve the quality of prompts for LLMs. The study suggests that because LLMs exhibit an impressive ability and quality in understanding natural language and in carrying out tasks in various domains such as answering questions, mathematical reasoning, code generation, translation and summarization of tasks, etc., a principled approach to curate various prompt types and techniques immensely improves a generated response.

Take this simple example of a prompt before and after applying one of the principles of conciseness and precision in the prompt. The difference in the response suggests both readability and simplicity—and the desired target audience for the response.

Press enter or click to view image in full size

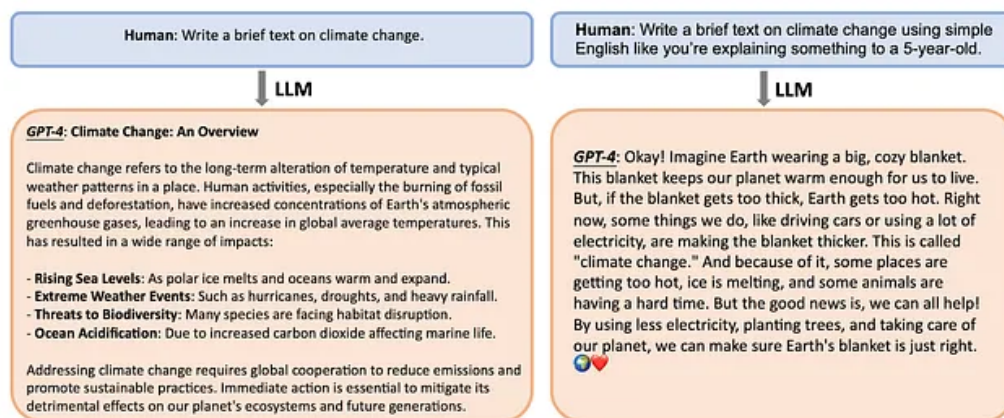


Figure 1: An Illustration showing a general prompt on the left and a precise and specific prompt on the right.[4]

As you might infer from the above example that the more concise and precise your prompt, the better LLM comprehends the task at hand, and, hence, the better response it formulates.

Let's examine some of the principles, techniques and types of prompt that offer better insight in how to carry out a task in various domains of natural language processing.

Prompt Principles and Guides

Bsharat et.al tabulate 26 ordered prompt principles, which can further be categorized into five distinct categories, as shown in Figure 2:

1. **Prompt Structure and Clarity:** Integrate the intended audience in the prompt.
2. **Specificity and Information:** Implement example-driven prompting (Use few-shot